

VibView

TikTok Profile Viewer & Downloader

Technical Documentation

Next.js 16

TypeScript

Tailwind CSS v4

Vercel

Upstash Redis

RapidAPI

Version 0.1.0 · Build Date: August 2026 · vibview-two.vercel.app

1. Project Overview

VibView is a web application that lets users anonymously browse any public TikTok profile — viewing videos, stories, highlights, reposts, and live streams — without needing a TikTok account. It also provides video/audio downloading and account analytics.

Live URL

vibview-two.vercel.app

Deployed on Vercel Hobby plan (free tier)

Version

0.1.0

Private repository, single-team project

Primary Language

TypeScript

Strict mode enabled, zero **any** policy

Architecture

Serverless

All backend logic runs as Vercel Functions

Core Features

Feature	Description
Profile Viewer	View any public TikTok profile with stats, follower count, bio
Video Grid	Infinite-scroll grid of all public videos with view counts
Stories Tab	View and download TikTok stories (discontinued in 2023)
Highlights Tab	Top 12 videos sorted by view count
Reposts Tab	Videos the user has reposted
Live Tab	Check if user is live; record up to 5 minutes of stream
Age Tab	Account age estimator based on earliest post date
Download	Download video (with/without watermark), audio, cover image
My Lists	Save and revisit favourite TikTok profiles
Dark / Light Mode	System-aware theme toggle

2. Technology Stack

Frontend

Layer	Technology	Version	Purpose
Framework	Next.js	16.3.2	App Router, React Server Components, API Routes
Runtime	React	19.2.8	UI rendering
Language	TypeScript	^5	Static typing, strict mode
Styling	Tailwind CSS	v4	Utility-first CSS, dark mode
Components	Base UI	^1.7.0	Headless accessible components (Radix-based)
Icons	Lucide React	^1.33.0	SVG icon library
Theming	next-themes	^0.4.6	Light / dark mode persistence
Video	HLS.js	^1.7.1	HTTP Live Streaming for TikTok live streams
Animations	tw-animate-css	^1.4.0	Tailwind animation utilities
Class utils	clsx + tailwind-merge	latest	Conditional class merging

Backend (Serverless API Routes)

Purpose	Technology	Notes
API layer	Next.js Route Handlers	Serverless functions, runs on Vercel
Schema validation	Zod	^4.4.3 — validates all provider API responses
Email	Resend	Contact form — 100 emails/day free
Rate limiting	@upstash/ratelimit	^2.0.8 — per-IP limits on API routes
Image proxy	Custom <code>/api/img</code>	Proxies TikTok CDN images (avoids CORS)
Download proxy	Custom <code>/api/dl</code>	Streams TikTok CDN video/audio to browser

Data & Caching

Purpose	Technology	Free Tier
Cache store	Upstash Redis	10,000 commands/day
Database	Supabase (PostgreSQL)	500 MB storage, 2 GB bandwidth/month

Cache TTLs

Bucket	TTL	Contents
<code>vv:user:{username}</code>	24 hours	UserProfile (followers, bio, avatar, etc.)
<code>vv:posts:{username}:{cursor}</code>	6 hours	Paginated video list per cursor position
<code>vv:video:{id}</code>	24 hours	Single video detail (download URLs, stats)
<code>vv:uid:{username}</code>	7 days	Numeric TikTok user ID (for ScrapTik)

3. TikTok Data Provider Chain

Because TikTok aggressively blocks datacenter IP addresses, no single provider is reliable. VibView uses a waterfall chain — each provider is tried in order; if one fails or times out (12 s), the next is attempted automatically.



#	Provider	Endpoint	Used For	Status
1	ScrapTik	scraptik.p.rapidapi.com	Video lists (getUserPosts)	✔ Working
2	TikHub	api.tikhub.io	Profile data (getUser)	✔ Working (free profile endpoint)
3	tikwm	tikwm.com/api	General fallback	✗ Cloudflare blocks Vercel IPs
4	tobyg74	tiktok.com/api/post/item_list	General fallback	✗ TikTok blocks Vercel IPs
5	HTML Scrape	tiktok.com/@username	Profile info last resort	⚠ Profile only, no video list

ScrapTik — Two-Step Video Fetch

ScrapTik requires a numeric user ID (not username). The provider resolves this automatically:

```
Step 1 – Resolve username → numeric ID (cached 7 days in Redis)
GET /username-to-id?username=kainattariq20
← { uid: "7008594880457344001", sec_uid: "MS4wLjAB..." }

Step 2 – Fetch video list
GET /user-posts?user_id=7008594880457344001&count=30&max_cursor=0@ion=US
← { aweme_list: [...], has_more: 1, max_cursor: 1786959737566 }
```

4. System Architecture

BROWSER

```

|
|— GET /@username
|   └─ Next.js Server Component (profile/[username]/page.tsx)
|       └─ tiktok.getUser() → Provider Chain → Redis cache (24h)
|           └─ tiktok.getUserPosts() → Provider Chain → Redis cache (6h)
|               └─ SSR: ProfileHeader + AnalyticsPanel + ProfileTabs
|
|— Client tab click (Stories / Highlights / Reposts / Live / Age)
|   └─ fetch /api/user/[username]/stories|reposts|live|posts
|       └─ Route Handler → Provider Chain → Response
|
|— Infinite scroll (load more videos)
|   └─ fetch /api/user/[username]/posts?cursor=1786959737566
|       └─ Route Handler → Provider Chain (ScrapTik max_cursor)
|
|— Image display
|   └─ /api/img?url=https://tiktokcdn-us.com/...jpeg
|       └─ Fetch CDN server-side → stream to browser (avoids CORS)
|
|— Video download
|   └─ /api/dl?url=https://tiktokv.com/...&dl=1
|       └─ Fetch CDN server-side → stream with Content-Disposition header

```

5. Key Files & Directory Structure

```

vibview/
├─ src/
│   ├─ app/
│   │   └─ api/
│   │       ├─ dl/route.ts           ← Video/audio download proxy
│   │       ├─ img/route.ts         ← Image proxy (TikTok CDN)
│   │       └─ user/[username]/
│   │           ├─ route.ts         ← GET user profile
│   │           ├─ posts/route.ts   ← GET paginated videos
│   │           ├─ stories/route.ts ← GET stories
│   │           ├─ reposts/route.ts ← GET reposts
│   │           └─ live/route.ts    ← GET live stream status
│   │               └─ video/[id]/route.ts ← GET single video detail
│   │   └─ profile/[username]/page.tsx ← Profile page (SSR)
│   │   └─ video/[id]/page.tsx      ← Video detail page
│   │   └─ layout.tsx               ← Root layout (Header, Footer)
│   │
│   └─ components/
│       └─ layout/
│           ├─ Header.tsx           ← Navigation bar
│           ├─ Footer.tsx          ← Site footer
│           └─ MyListsDrawer.tsx    ← Saved profiles sidebar
│       └─ profile/
│           ├─ ProfileTabs.tsx      ← Videos/Stories/Highlights/Reposts/Live/Age
│           ├─ ProfileHeader.tsx    ← Avatar, bio, follower stats
│           ├─ VideoGrid.tsx        ← Infinite-scroll video grid
│           ├─ AnalyticsPanel.tsx   ← Engagement metrics
│           └─ GrowthChart.tsx      ← Historical growth chart
│       └─ ui/                      ← Reusable UI components (Button, Card, Tabs ...)
│   └─ lib/
│       └─ providers/
│           ├─ chain.ts             ← Waterfall orchestrator + Redis caching
│           ├─ types.ts             ← Shared TypeScript types + Zod schemas
│           ├─ scraptik.ts          ← ScrapTik (RapidAPI) provider
│           ├─ tikhub.ts            ← TikHub provider
│           ├─ tikwm.ts             ← tikwm provider (blocked, kept as fallback)
│           ├─ tobyg74.ts           ← tobyg74 provider (blocked, kept as fallback)
│           └─ scrape.ts            ← HTML scrape provider
│       └─ cache.ts                 ← Upstash Redis wrapper
│   └─ env.ts                       ← All environment variable declarations
├─ package.json
└─ next.config.ts

```

6. API Endpoints

Method	Path	Description
GET	<code>/api/user/[username]</code>	Returns UserProfile JSON
GET	<code>/api/user/[username]/posts?cursor=0</code>	Returns paginated PostPage JSON
GET	<code>/api/user/[username]/stories</code>	Returns Story[] JSON
GET	<code>/api/user/[username]/reposts</code>	Returns PostPage JSON
GET	<code>/api/user/[username]/live</code>	Returns LiveRoom JSON or 404
GET	<code>/api/video/[id]</code>	Returns VideoDetail JSON
GET	<code>/api/img?url=...</code>	Proxies & serves TikTok CDN image
GET	<code>/api/dl?url=...&dl=1</code>	Streams video/audio download
POST	<code>/api/contact</code>	Sends contact form email via Resend

7. Environment Variables

Variable	Required	Free Tier	Purpose
<code>RAPIDAPI_KEY</code>	Yes	100 calls/month	ScrapTik authentication
<code>UPSTASH_REDIS_REST_URL</code>	Yes	10k cmds/day	Redis cache endpoint
<code>UPSTASH_REDIS_REST_TOKEN</code>	Yes	Free	Redis auth token
<code>TIKHUB_API_KEY</code>	Optional	Free profile endpoints	TikHub fallback provider
<code>SUPABASE_URL</code>	Optional	500 MB	My Lists database
<code>SUPABASE_ANON_KEY</code>	Optional	Free	Supabase auth
<code>RESEND_API_KEY</code>	Optional	100/day	Contact form emails
<code>TIKWM_API_TOKEN</code>	No	—	tikwm (currently blocked)
<code>PROXY_BASE_URL</code>	No	—	CF Worker URL (deprecated)
<code>PROXY_HMAC_SECRET</code>	No	—	CF Worker HMAC (deprecated)

8. Free Tier Limits & Sustainability

Service	Free Limit	Impact
ScrapTik (RapidAPI)	100 API calls/month	⚠️ Main bottleneck. ~50 unique profile loads/month (username→ID + posts). Repeat visits use cached ID = 1 call each.
Vercel	100 GB bandwidth/month	Video streaming uses bandwidth. A typical TikTok video is 5–20 MB. ~5,000–20,000 downloads/month before limit.
Upstash Redis	10,000 commands/day	✅ Comfortable. Each page load = ~4 Redis ops. Supports ~2,500 uncached page loads/day.
Supabase	500 MB storage, 2 GB bandwidth	✅ Fine for My Lists feature at any personal usage level.
Resend	100 emails/day	✅ Fine for contact form.

Upgrade Path

When ScrapTik's 100-call/month free limit is reached, upgrade to the **Basic plan (~\$10/month)** for 10,000 calls/month — sufficient for a fully public application.

9. Deployment

The app is deployed to **Vercel** using the Vercel CLI:

```
npm install -g vercel      # Install CLI
vercel --prod              # Deploy to production (from project root)
vercel env pull .env.local # Pull environment variables locally
```

Live URL: <https://vibview-two.vercel.app>

Project: [jabbar-alams-projects/vibview](#) on Vercel dashboard

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